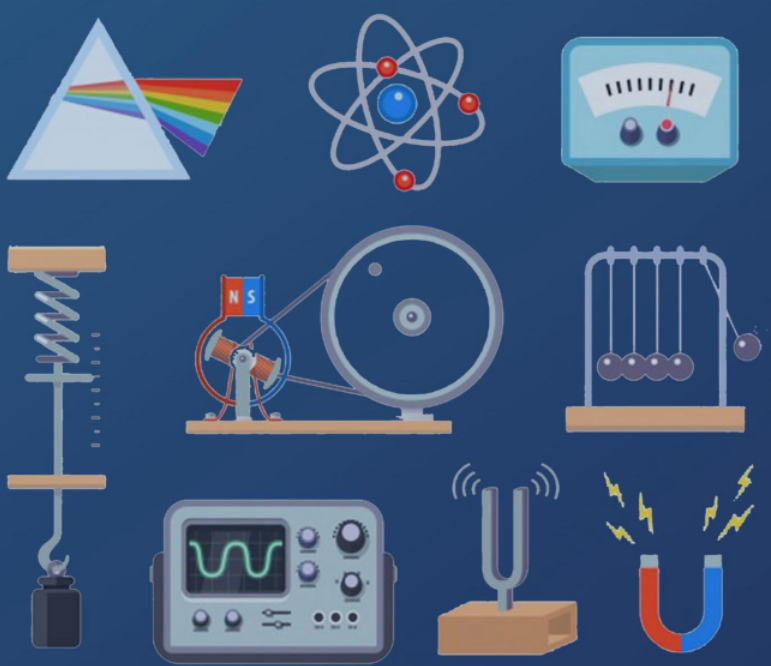
PHYSICS

Kinematics 1D



SHEET

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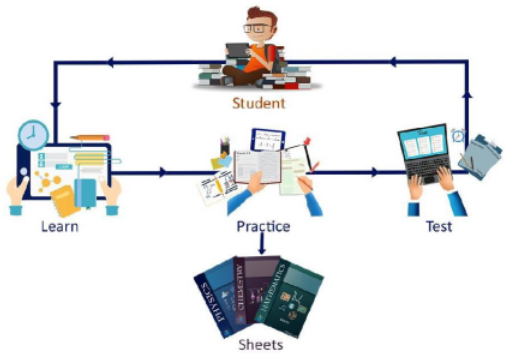
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Distance and Displacement	1, 2, 4			
2	Differentiation	5, 6, 7, 8, 9	1	2, 3, 4	
3	Speed, Velocity and Acceleration	3, 10-12	2, 3, 5	1, 5, 6	1
4	Motion Under Constant Acceleration-Part1	14	6, 18	8	
5	Motion Under Constant Acceleration-Part2	13, 15	12, 16, 17	7, 9, 10	12
6	Motion Under Gravity-Part1	16, 18	15		
7	Motion Under Gravity-Part2	17, 19	4, 7, 21	11, 12, 15, 16, 13, 14	11, 13
8	Relative Motion-Part1	20, 21, 22	19		
9	Relative Motion-Part2	23		17	15
10	Graphical Interpretation of Differentiation-Part1	24, 25, 26, 27	8, 9	18, 19	
11	Graphical Interpretation of Differentiation-Part2	28, 29, 30	10, 13		
12	Graphs	31, 33, 34			2
13	Conversion of Graphs	32	11, 14		3
14	Integration-Part1	36, 37	20	20	
15	Integration-Part2	35		21	5, 14
16	Integration-Part3	38		21	4, 6, 7, 8, 9, 10

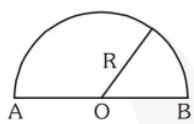
Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Distance and Displacement

Q1. A particle moves from A to B in a semicircle. Then the ratio of distance to displacement is :-



(A) $\frac{\pi}{2}$ (B) $\frac{2}{\pi}$

(C) $\frac{\pi}{4}$ (D) 1 : 1

Q2. A table clock has minute hand 4 cm long. The displacement of tip of minute hand between 6:00 pm to 6:15 pm is :-

(A) 4 cm (B) πcm

(C) $4\sqrt{2}$ cm (D) 8 cm

Q3. A man walks 5m north then 4m east then $5\sqrt{2}$ m south-west. The displacement from initial to final position is :-

(A) 1 m West

(B) 1m East

(C) $10\sqrt{2}$ m North-east

(D) None of above

Rate of Change Or Differentiation

Q4. $\frac{d}{dx}\left(\frac{1}{x^7}\right)$ is -

(A) $-7x^{-6}$ (B) $-7x^{-8}$

(C) $7x^{-6}$ (D) $7x^{-8}$

Q5. If $v = t^2 - 4t + 5$ then $\frac{dv}{dt}$ is -

(A) $2t - 4$ (B) $t - 4$

(C) $2t^2 + 5$ (D) Zero

Q6. $\frac{d}{d\theta}(\sin 2\theta)$ is

(A) $2\cos 2\theta$ (B) $\frac{1}{2}\cos 2\theta$

(C) $-2\cos 2\theta$ (D) $-\frac{1}{2}\cos 2\theta$

Q7. The value of $\frac{d}{dx}\left(\frac{x}{\sin x}\right)$ will be :-

(A) $\frac{\sin x - x\cos x}{\sin^2 x}$ (B) $\frac{\cos x - x\sin x}{\sin^2 x}$

(C) $\frac{x\cos x - \sin x}{\cos^2 x}$ (D) $\frac{x\sin x - \cos x}{\cos^2 x}$

Q8. $\frac{d}{dx}\left(e^{\sin\frac{\pi}{4}}\right)$ is -

(A) $e^{\cos\frac{\pi}{4}}$ (B) $\frac{\pi}{4}e^{\sin\frac{\pi}{4}}$

(C) $\frac{\pi}{4}e^{\sin\frac{\pi}{4}}\cos\frac{\pi}{4}$ (D) Zero

Speed, Velocity and Acceleration

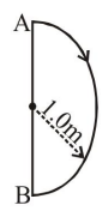
Q9. In 1.0 sec. a particle goes from point A to point B moving in a semicircle of radius 1.0 m. The magnitude of average velocity is:-

(A) 3.14 m/sec

(B) 2.0 m/sec

(C) 1.0 m/sec

(D) zero



Q10. A particle moves along x-axis in such a way that its co-ordinate x varies with time t according to the equation $x = (2 - 5t + 6t^2)$ m. The initial velocity of the particle is :

(A) – 5 m/s (B) 6 m/s

(C) – 3 m/s (D) 3 m/s

Q11. The displacement x of a particle along a straight line at time t is given by

$x = a_0 + a_1t + a_2t^2$.

The acceleration of the particle is:

(A) a_0 (B) a_1

(C) $2a_2$ (D) a_2

Q12. The velocity of a particle moving on the x-axis is given by $v = x^2 + x$ where v is in m/s and x is in m. Find its acceleration in m/s^2 when passing through the point x = 2m.

(A) 0 (B) 5

(C) 11 (D) 30

Motion Under Constant Acceleration

Q13. A car starts from rest travelling with constant acceleration. If distance covered by it in 10th second of its journey is 19m, what will be the acceleration of car :-

(A) $4\ m/s^2$ (B) $3\ m/s^2$

(C) $2\ m/s^2$ (D) $1\ m/s^2$

Q14. A body starts from rest and is uniformly accelerated for 30 s. The distance travelled in the first 10 s is x_1, next 10 s is x_2 and the last 10 s is x_3. Then $x_1 : x_2 : x_3$ is the same as:-

(A) 1 : 2 : 4 (B) 1 : 2 : 5

(C) 1 : 3 : 5 (D) 1 : 3 : 9

Q15. A car moving with a speed of 12 m/s can be stopped by applying brakes after travelling a distance of 6m. If the same car is moving with a speed of 24 m/s then the stopping distance of the car is :

(A) 24 m (B) 12 m

(C) 18 m (D) 30 m

Motion Under Gravity part

Q16. A ball is projected upwards. Its acceleration at the highest point is :-

(A) zero

(B) directed upward

(C) directed downwards

(D) such as cannot be predicted

Q17. A body is thrown upward and reaches its maximum height. At that position-

(A) Its velocity is zero and its acceleration is also zero.

(B) Its velocity is zero but its acceleration is maximum.

(C) Its acceleration is minimum.

(D) Its velocity is zero and its acceleration is acceleration due to gravity.

Q18. A stone dropped from the top of the tower touches the ground in 4 sec. The height of the tower is about :-

(A) 80 m (B) 40 m

(C) 20 m (D) 160 m

Q19. With what speed should a body be thrown upwards so that the distances traversed in 5th second and 6th second are equal ?

(Take $g = 9.8\ m/s^2$)

(A) 58.4 m/s (B) 49 m/s

(C) $\sqrt{98}$ m/s (D) 98 m/s

Relative Motion

Q20. Two cars are moving towards each other with velocity 4m/s & 8 m/s respectively from two different points in straight line, 1200 m apart simultaneously, find time after which both cars meet :-

(A) 300 sec

(B) 150 sec

(C) 100 sec

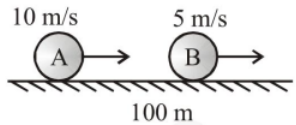
(D) None of these

Q21. Two cars A & B start from rest (from the same point) in same direction with acceleration 8 m/s² & 4 m/s² respectively then acceleration of car B in frame of A (Take direction of motion of car is positive):-

(A) 4 m/s² (B) – 4 m/s²

(C) 12 m/s² (D) None of these

Q22. An object A is moving with 10 m/s and B is moving with 5 m/s in the same direction of positive x- axis. A is 100 m behind B as shown. Find time taken by A to meet B :-



(A) 18 sec (B) 16 sec

(C) 20 sec (D) 17 sec

Q23. If a lift is moving downwards with constant velocity 5 m/s. A coin is dropped from the lift, find velocity of coin at the same instant when it is dropped in frame of lift and in frame of ground :-

(A) 5 m/s (downward) & 0

(B) 0 & 5 m/s (downward)

(C) 5 m/s (upward) & 0

(D) 0 & 5 m/s (upward)

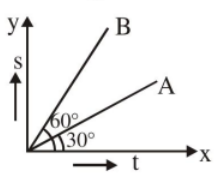
Graphical Interpretation of Differentiation

Q24. Slope of tangent to velocity time curve is :-

(A) Distance (B) Speed

(C) Acceleration (D) None of above

Q25. The displacement-time graphs of two bodies A and B are shown in figure. The ratio of velocity of A, v_A to velocity of B, v_B is :-



(A) $\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $\frac{1}{3}$ (D) 3

Q26. The slope of the velocity-time graph for retarded motion is :-

(A) zero

(B) positive

(C) negative

(D) positive or negative

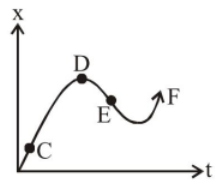
Q27. The displacement–time graph of a moving particle is shown below. The instantaneous velocity of the particle is negative at the point.

(A) C

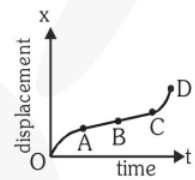
(B) D

(C) E

(D) F



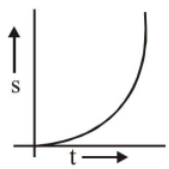
Q28. The graph between the displacement x and time t for a particle moving in a straight line is shown in figure. During the interval OA, AB, BC and CD, the acceleration of the particle is :



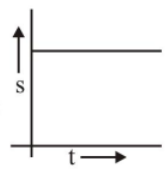
	OA,	AB,	BC,	CD
(A)	+	0	+	+
(B)	–	0	+	0
(C)	+	0	–	+
(D)	–	0	0	+

Q29. A body is travelling in a straight line with a uniformly increasing speed. Which one of the plot represents the distance (s) travelled with time (t) ?

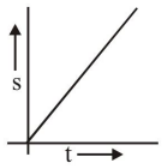
(A)



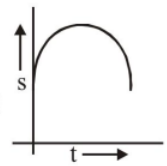
(B)



(C)



(D)



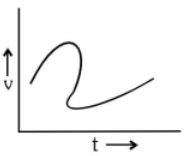
Q30. If $y = x^3 - 12x + 4$ then condition for maximum y is -

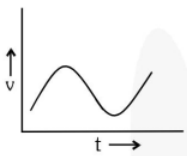
(A) $x = 2$ (B) $x = -2$

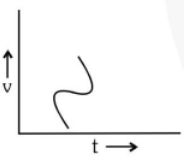
(C) $x = \pm 2$ (D) $x = 4$

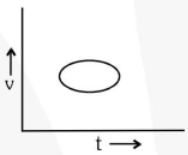
Graphs

Q31. Which of the following velocity–time graph shows a realistic situation for a body in motion :–

(A) 

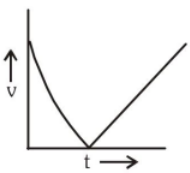
(B) 

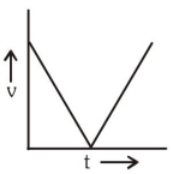
(C) 

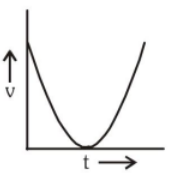
(D) 

Q32. A ball is thrown vertically upwards. Which of the following plots represents the speed–time graph of the ball during its flight if air resistance is ignored :–

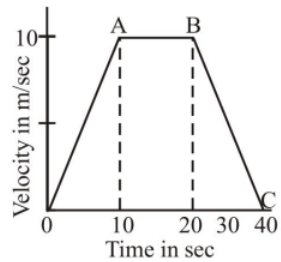
(A) 

(B) 

(C) 

(D) 

Q33. The adjoining curve represents the velocity–time graph of a particle, its acceleration values along OA, AB and BC in metre/sec^2 are respectively :-

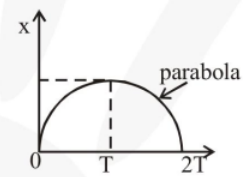


(A) 1, 0, –0.5 (B) 1, 0, 0.5

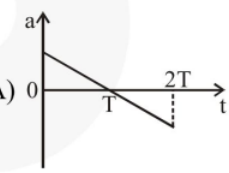
(C) 1, 1, 0.5 (D) 1, 0.5, 0

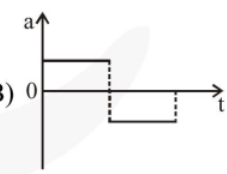
Conversion of Graphs

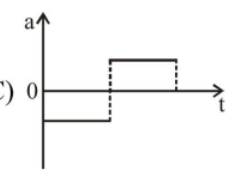
Q34. The x–t graph of a particle moving along a straight line is shown in figure.

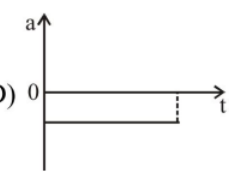


The a–t graph of the particle is correctly shown by :-

(A) 

(B) 

(C) 

(D) 

Integration

Q35. The value of $\int \frac{1}{x^3}\,dx$ is -

(A) $-3x^{-4}+c$ (B) $\frac{x^{-4}}{-4}+c$

(C) $\frac{x^{-2}}{-2}+c$ (D) $-2x^{-2}+c$

Q36. The value of $\int_0^1 (x^2-4x+5)\,dx$ is :-

(A) $\frac{5}{3}$ (B) $\frac{10}{3}$

(C) $\frac{8}{3}$ (D) $\frac{11}{3}$

Q37. The area under acceleration-time graph gives:-

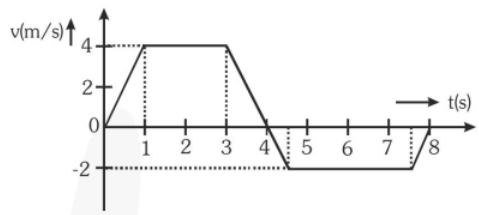
(A) Distance travelled

(B) Change in acceleration

(C) Force acting

(D) Change in velocity

Q38. The velocity – time graph of a linear motion is shown in figure. The displacement & distance from the origin after 8 sec. is :-



(A) 5 m, 19m (B) 16 m,22m

(C) 8 m,19m (D) 6 m, 5m

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Mark the correct options

if $x=\frac{1}{t^{3/8}}$, $y=t^{-4}$:-

(A) $\frac{dx}{dt}=-\frac{3}{8}t^{-5/8}$ (B) $\frac{dx}{dt}=-\frac{3}{8}t^{-11/8}$

(C) $\frac{dy}{dt}=\frac{4}{t^5}$ (D) $\frac{dy}{dt}=-4t^{-5}$

Q2. The position of a bike is given by $x = t^2$. Mark the correct option :-

(A) Average speed between 2 to 3 second is 5 m/s

(B) Average speed between 2 to 3 second is 4 m/s

(C) Average speed between 2 to 2.1 second is 4.1 m/s

(D) Average speed between 2 to 2.1 second is 4.01 m/s

Q3. The position vector of a particle is given as $\vec{r}=e^t\hat{i}+\log_e t\,\hat{j}$. Find $\vec{v}$ and $\vec{a}$ at $t = 2$sec.

(A) $\vec{v}=\left(e^2\hat{i}+\frac{1}{2}\hat{j}\right)m/s$

(B) $\vec{a}=\left(e^2\hat{i}-\frac{1}{4}\hat{j}\right)m/s^2$

(C) $\vec{v}=\left(e^2\hat{j}+1\hat{k}\right)m/s$

(D) $\vec{a}=\left(e^2\hat{i}+\frac{1}{4}\hat{j}\right)m/s^2$

Q4. Which of the following statements are true for a moving body ?

(A) If its speed changes, its velocity must change and it must have some acceleration.

(B) If its velocity changes, its speeds must change and it must have some acceleration.

(C) If its velocity changes, its speed may or may not change, and it must have some acceleration.

(D) If its speed changes but direction of motion does not change, its velocity may remain constant.

Q5. The numerical ratio of magnitude of average velocity to average speed can be :-

(A) equal to one

(B) less than one and greator than zero

(C) zero

(D) more than one

Q6. A train accelerates from rest for time t_1 at a constant acceleration α for distance x. Then it decelerates to rest at constant retardation β in time t_2 for distance y. Then -

(A) $\frac{x}{y}=\frac{\beta}{\alpha}$ (B) $\frac{\beta}{\alpha}=\frac{t_1}{t_2}$

(C) $\frac{x}{y}=\frac{t_1}{t_2}$ (D) $x = y$

Q7. An object may have :-

(A) Varying speed without having varying acceleration

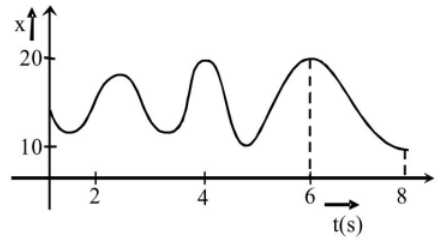
(B) Varying velocity without having varying speed

(C) Non–zero acceleration without having varying velocity

(D) Non–zero acceleration without having varying speed .

Q8. In the figure is shown the position of a particle moving on the x-axis as a function of time.

Then -



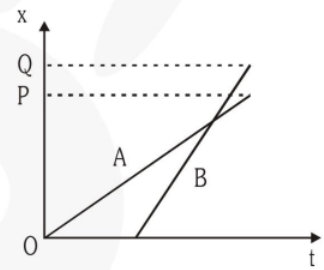
(A) the particle has come to rest for 6 times

(B) the maximum speed is at t = 6 s

(C) the velocity remain positive for t = 0 to t = 6s

(D) the average velocity for the total period shown is negative

Q9. The position–time (x–t) graphs for two children A and B returning from their school O to their homes P and Q respectively are shown in fig. They travel on same road. Choose the correct options:



(A) A lives closer to the school than B

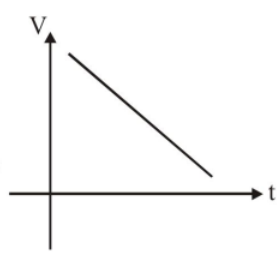
(B) B starts from the school earlier than A

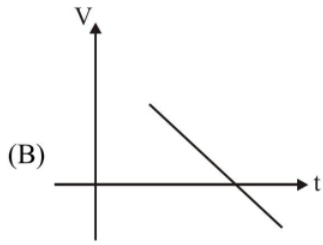
(C) B walks faster than A

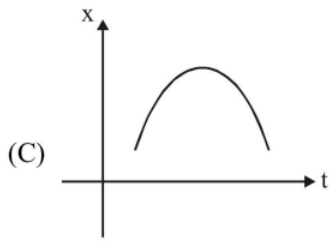
(D) B overtakes A on the road once.

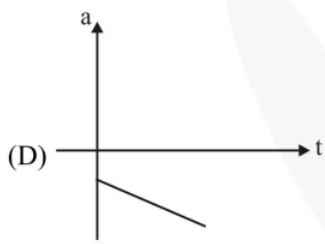
Q10. Mark correct option in which acceleration is negative :-

(A)

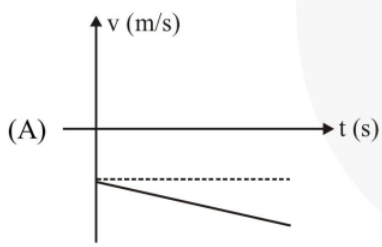


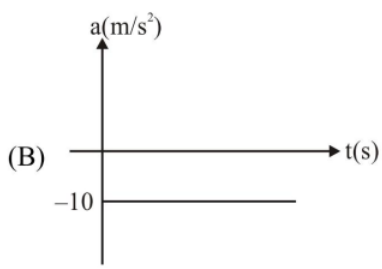


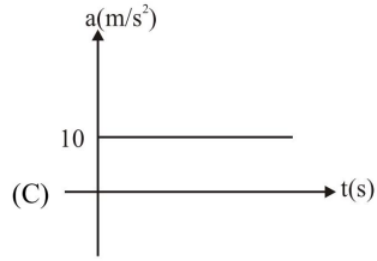


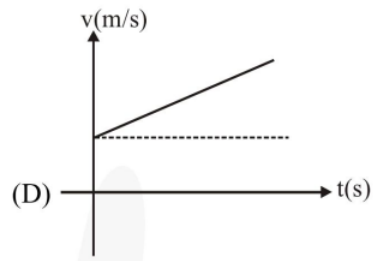


Q11. A particle is projected from top of a tower in vertical downward direction with velocity 10 m/s. Taking upward direction as +ve, choose plot a-t and v-t graph :-
[Take g = 10m/s^2]









MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

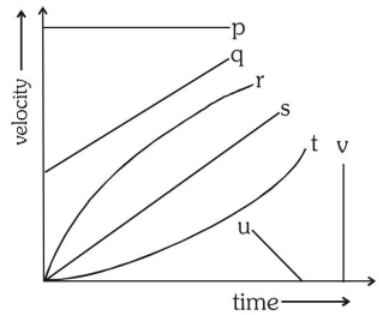
Q12. The equation of one dimensional motion of the particle is described in column I. At t = 0, particle is at origin and at rest. Match the column I with the statements in Column II.

Column I	**Column II**
(A) $x = (3t^2 + 2)$m	(p) Velocity of particle at t = 1 s is 8 m/s
(B) v = 8t m/s	(q) Particle moves with uniform acceleration
(C) a = 16t	(r) Particle moves with variable acceleration
(D) $v = 6t - 3t^2$	(s) Particle will change its direction some time.

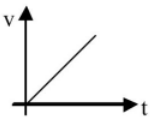
Q13. **Column I**

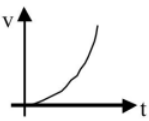
(A) Zero acceleration
(B) Infinite acceleration
(C) Constant positive acceleration, with zero initial velocity
(D) Constant positive acceleration, with non–zero initial velocity
(E) Constant Negative acceleration
(F) Increasing acceleration
(G) Decreasing acceleration

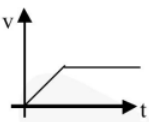
Column II

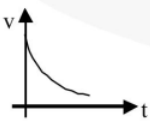


Q14. **Column I**

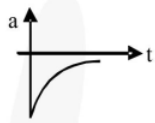
(A) 

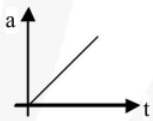
(B) 

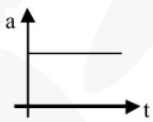
(C) 

(D) 

Column II

(p) 

(q) 

(r) 

(s) 

ASSERTION AND REASON

Q15. **Statement I :** When velocity of a particle is zero then acceleration of particle is zero.

Statement II : Acceleration is equal to rate of change of velocity.

(A) Statement–I is true, Statement–II is true; Statement–II is correct explanation for Statement–I.
(B) Statement–I is true, Statement–II is true; Statement–II is NOT a correct explanation for statement–I.
(C) Statement–I is true, Statement–II is false.
(D) Statement–I is false, Statement–II is true.

Q16. **Statement I :** A particle moves in a straight line with constant acceleration. The average velocity of this particle cannot be zero in any time interval.

Statement II : For a particle moving in straight line with constant acceleration, the average velocity in a time interval is $\frac{u+v}{2}$, where u and v are initial and final velocity of the particle of the given time interval.

(A) Statement–I is true, Statement–II is true; Statement–II is correct explanation for Statement–I.
(B) Statement–I is true, Statement–II is true; Statement–II is NOT a correct explanation for statement–I.

(C) Statement–I is true, Statement–II is false.

(D) Statement–I is false, Statement–II is true.

Q17. **Statement–I :** A positive acceleration can be associated with a 'slowing down' of the body.

Statement–II : The origin and the positive direction of an axis are a matter of choice.

(A) Statement–I is true, Statement–II is true; Statement–II is correct explanation for Statement–I.

(B) Statement–I is true, Statement–II is true; Statement–II is NOT a correct explanation for statement–I.

(C) Statement–I is true, Statement–II is false.

(D) Statement–I is false, Statement–II is true.

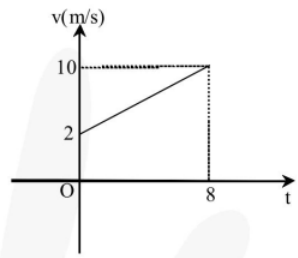
NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. A driver travelling at speed 36 kmh^{-1} sees the light turn red at the intersection. If his reaction time is 0.6s, and then the car can deaccelerate at $4ms^{-2}$. Find the stopping distance of the car.

Q19. Two cars travelling towards each other on a straight road at velocity 10 m/s and 12 m/s respectively. When they are 150 metre apart, both drivers apply their brakes and each car decelerates at 2 m/s^2 until it stops. How far apart will they be when they have both come to a stop ?

Q20. Figure shows the graph of velocity versus time for a particle going along x axis. Initially at t = 0, particle is at x = 3m. Find position of particle at t = 2s. (in m).



SUBJECTIVE TYPE

Q21. A ball is thrown vertically upwards with a velocity of 20 ms^{-1} from the top of a tower. The height of the tower is 25 m from the ground.

(i) How high will the ball rise from the top of the tower?

(ii) How long will it be before the ball hits the ground ? (Take g = 10 ms^{-2})

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If $y = (x^2 + 4x - 5)^3$ then $\frac{dy}{dx}$ is :-

(A) $3\ (x^2 + 4x - 5)^2$

(B) $3\ (x^2 + 4x - 5)^4$

(C) $3\ (x^2 + 4x - 5)^3\ (2x + 4)$

(D) $3\ (x^2 + 4x - 5)^2\ (2x + 4)$

Q2. Temperature of a body varies with time as $T = T_0 + \alpha t^2 + \beta \sin t)$ where T_0 is the temperature in Kelvin at $t = 0$ sec. & $\alpha = 2/\pi$ K/s^2 & $\beta = -4$ K, then rate of change of temperature at $t = \pi$ sec. is-

(A) 8 K (B) 8K

(C) 8K/sec (D) 8K/sec

Q3. If $x = at^2$ & $y = bt^3$, then $\frac{dy}{dx} = ?$

(A) $\frac{3bt}{2a}$ (B) $\frac{2bt}{3a}$ (C) $\frac{2at}{3b}$ (D) $\frac{3b}{2a}$

Q4. A body has speed V, 2V and 3V for first 1/3rd of the distance S, second 1/3rd of S and third 1/3rd of S respectively. Its average speed will be :-

(A) V (B) 2V

(C) $\frac{18}{11}$V (D) $\frac{11}{18}$V

Q5. The velocity of a particle moving along x–axis is given as $v = x^2 - 5x + 4$ (in m/s) where x denotes the x–coordinate of the particle in metres. Find the magnitude of acceleration of the particle when the velocity of particle is zero ?

(A) 0 m/s^2 (B) 2 m/s^2

(C) 3 m/s^2 (D) None of these

Q6. The relation between time t and distance x is $t = \alpha x^2 + \beta x$ where α and β are constants. The retardation is :-

(A) $2\alpha v^3$ (B) $2\beta v^2$

(C) $2\alpha\beta v^2$ (D) $2\beta^2 v^3$

Q7. A particle is moving with uniform acceleration along a straight line. Its velocities at A & B are respectively 7 m/s & 17 m/s . M is mid point of AB . If t_1 is the time taken to go from A to M and t_2 the time taken to go from M to B, the ratio $\frac{t_1}{t_2}$ is equal to :-

(A) 3 : 2 (B) 3 : 1

(C) 2 : 1 (D) 2 : 3

Q8. A particle moves along a straight line OX. At a time t (in seconds) the distance x (in metres) of the particle from O is given by $x = 40 + 12t - t^3$. How much distance would the particle travel before coming to rest ?

(A) 24 m (B) 40 m

(C) 56 m (D) 16 m

Q9. A particle moves with uniform acceleration and v_1, v_2 and v_3 denote the average velocities in the three successive intervals of time t_1, t_2 and t_3. Which of the following relations is correct ?

(A) $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_2 + t_3)$

(B) $(v_1 - v_2) : (v_2 - v_3) = (t_1 + t_2) : (t_2 + t_3)$

(C) $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_1 - t_3)$

(D) $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_2 - t_3)$

Q10. The engine of a motorcycle can produce a maximum acceleration 5 m/s^2. Its brakes can produce a maximum retardation 10 m/s^2. If motorcyclist start from point A and reach at point B. What is the minimum time in which it can cover if distance between A and B is 1.5 km: Given that motorcycle comes to rest at B.

(A) 30 sec (B) 15 sec

(C) 10 sec (D) 5 sec

Q11. Balls are thrown vertically upward in such a way that the next ball is thrown when the previous one is at the maximum height. If the maximum height is 5m, the number of balls thrown per minute will be :-

(A) 40 (B) 50

(C) 60 (D) 120

Q12. A parachutist drops freely from an aeroplane for 10 s before the parachute opens out. Then he descends with a net retardation of 2.5 m/s^2. If he bails out of the plane at a height of 2495 m and g = 10 m/s^2, hit velocity on reaching the ground will be :-

(A) 5 m/s (B) 10 m/s

(C) 15 m/s (D) 20 m/s

Q13. A lift is moving downward with constant velocity 10 m/s. A coin is dropped at height 5m from the floor of lift, find the time taken by coin to reach the floor :-

(A) 1 s (B) 0.5 s

(C) 2 s (D) 1.5 s

Q14. A lift is moving with velocity 10 m/s upward with acceleration 5 m/s^2 upwards. At the same instant of time, a coin is dropped from 7.5 m above the floor of lift, find distance travelled by the coin in ground frame before it strikes the floor :-

(A) 5 m (upward)

(B) 10 m (downward)

(C) 10 m (upward)

(D) 5 m (downward)

Q15. A, B, C and D are points in a vertical line such that AB = BC = CD. If a body falls from rest from A, then the times of descend through AB, BC and CD are in the ratio :-

(A) $1:\sqrt{3}:\sqrt{2}$

(B) $\sqrt{2}:\sqrt{3}:1$

(C) $\sqrt{3}:1:\sqrt{2}$

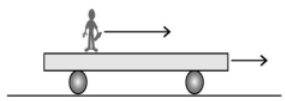
(D) $1:(\sqrt{2}-1):(\sqrt{3}-\sqrt{2})$

Q16. A particle is thrown upwards from ground. It experiences a constant resistance force which can produce retardation 2 m/s^2. The ratio of time of ascent to the time of descent is :- [g = 10 m/s^2]

(A) 1 : 1 (B) $\sqrt{\frac{2}{3}}$

(C) $\frac{2}{3}$ (D) $\sqrt{\frac{3}{2}}$

Q17. A trolley is moving horizontally with a constant velocity of v m/s w.r.t. earth. A man starts running from one end of the trolley with a velocity 1.5v m/s w.r.t. to trolley. After reaching the opposite end, the man return back and continues running with a velocity of 1.5 v m/s w.r.t. the trolley in the backward direction. If the length of the trolley is L then the displacement of the man with respect to earth during the process will be :-



(A) 2.5L (B) 1.5 L

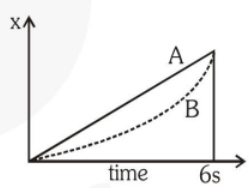
(C) $\frac{5L}{3}$ (D) $\frac{4L}{3}$

Q18. If force $F = \frac{Kq(Q-q)}{r^2}$ then condition for maximum force F is (K, Q and r are constants):-

(A) $q = \frac{Q}{4}$ (B) $q = \frac{Q}{2}$

(C) q = Q (D) q = 2Q

Q19. In the diagram shown, the displacement of particles is given as a function of time. The particle A is moving under constant velocity of 9 m/s. The particle B is moving under variable acceleration. From time t = 0 s. to t = 6 s., the average velocity of the particle B will be equal to :-



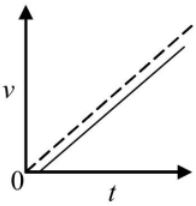
(A) 2.5 m/s (B) 4 m/s

(C) 9 m/s (D) None

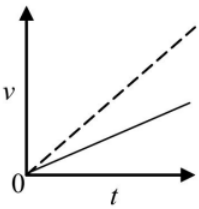
Q20. If an object is released in air, the magnitude of its acceleration would begin at the free-fall value, but it would decrease continuously to zero as the object continued to fall. For which one of the choices given does the solid line best represent the speed of the object as a function of time when it is dropped from rest in air?

Note: The dashed line shows the free-fall under vacuum graph for comparison.

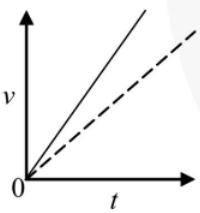
(A)



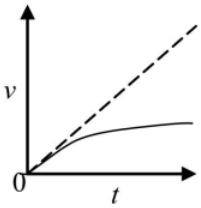
(B)



(C)



(D)



Q21. Acceleration of a particle is given by $a = 6x^2 - 4x + 5$ m/s^2. If the particle is initially at origin and moving with velocity of 20 m/s then velocity at x = 5m will be :-

(A) $\sqrt{450}$ m/s

(B) $\sqrt{650}$ m/s

(C) $\sqrt{850}$ m/s

(D) None of these

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A particle moves along the X–axis as $x = u(t-2) + a(t-2)^2$:-

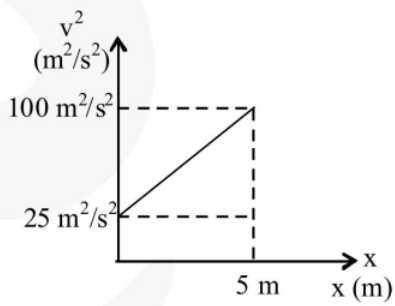
(A) The initial velocity of the particle is u

(B) The acceleration of the particle is a

(C) The acceleration of the particle is 2a

(D) At t = 2s particle is at the origin.

Q2. A particle is moving along x-axis and graph between square of speed and position of the particle is given in the figure. If at t = 0, x = 0 m, then select correct statement :-



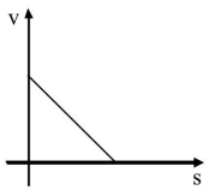
(A) Acceleration of the particle is 15 m/s^2 at $t = \frac{1}{2}$ s

(B) Acceleration of the particle is 7.5 m/s^2 at t = 1s

(C) Acceleration of the particle is constant

(D) At t = 1 s, velocity of particle is 12.5 m/s

Q3. Velocity-displacement graph of a particle moving in a straight line is as shown in figure :-



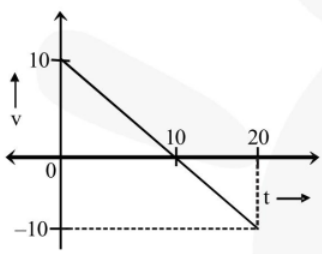
(A) magnitude of acceleration of particle is decreasing

(B) magnitude of acceleration of particle is increasing

(C) acceleration *versus* displacement graph is straight line

(D) acceleration *versus* displacement graph is parabola

Q4. The velocity-time plot for a particle moving on a straight line is shown in figure:-



(A) the particle has constant acceleration

(B) the particle has never turned around

(C) the particle has zero displacement

(D) the average speed in the interval 0 to 10 sec is same as the average speed in the interval 10 sec to 20 sec

Q5. A particle moves along a straight line and its velocity depends on time as $v = 4t - t^2$. Then for first five second :-

(A) Average velocity is 25/3 m/s

(B) Average speed is 10 m/s

(C) Average velocity is 5/3 m/s

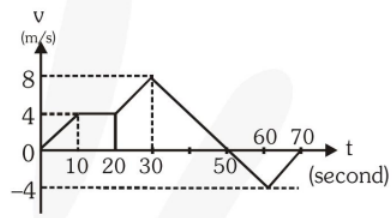
(D) Acceleration is 4 m/s^2 at t = 0

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

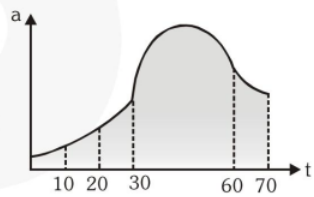
Paragraph for Question No. 6 to 10

A car is moving on a straight road. The velocity of the car varies with time as shown in the figure. Initially (at t = 0), the car was at x = 0, where, x is the position of the car at any time 't'.

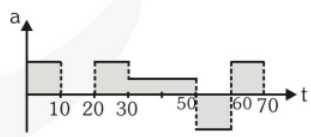


Q6. The variation of acceleration (a) with time (t) will be best represented by :-

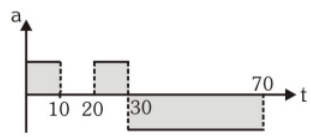
(A)



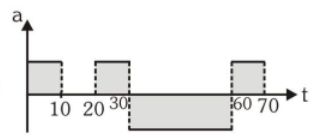
(B)



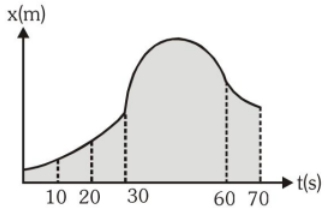
(C)

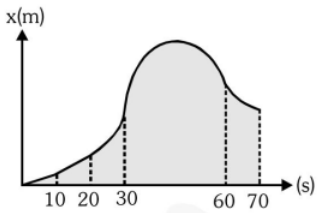


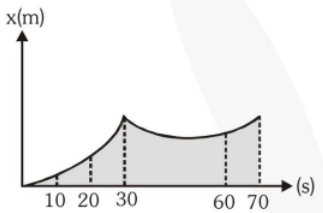
(D)

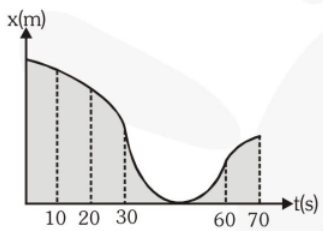


Q7. The displacement time graph will be best represented by :-

(A)


(B)


(C)


(D)


Q8. The maximum displacement from the starting position will be :-

(A) 200 m (B) 250 m

(C) 160 m (D) 165 m

Q9. Average speed from t = 0 to t = 70 s will be :-

(A) $\frac{16}{7}$ m/s (B) $\frac{24}{7}$ m/s

(C) $\frac{20}{7}$ m/s (D) zero

Q10. The time interval during which the car is retarding can be :-

(A) t = 50s to t = 70s

(B) t = 30s to t = 50s

(C) t = 30s to t = 60s

(D) t = 10s to t = 20s

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11. A balloon rises up with constant net acceleration of $10 m/s^2$. After 2 s a particle drops from the balloon. After further 2 s match the following : (Take g = 10 m/s^2)

Column I	**Column II**
(A) Height of particle from ground	(p) Zero
(B) Speed of particle	(q) 10 SI units
(C) Displacement of Particle	(r) 40 SI units
(D) Acceleration of particle	(s) 20 SI units

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q12. A train, travelling at 20 km/hr is approaching a platform. A bird is sitting on a pole on the platform. When the train is at a distance of 2 km from pole, brakes are applied which produce a uniform deceleration in it to stop it at pole. At that instant the bird flies towards the train at 60 km/hr and after touching the nearest point on the train flies back to the pole and then flies towards the train and continues repeating itself. Calculate how much distance will the bird have flown before the train stops?

SUBJECTIVE TYPE

Q13. A juggler performs in a room whose ceiling is 3 m above the level of his hands. He throws a ball vertically upward so that it just reaches the ceiling.

(a) With what initial velocity does he throw the ball ?

(b) What time is required for the ball to reach the ceiling ?

He throws a second ball upward with the same initial velocity, at the instant that the first ball is at the ceiling.

(c) How long after the second ball is thrown do the two ball pass each other ?

(d) When the balls pass each other, how far are they above the juggler's hands ?

Q14. A particle is moving along x – axis with acceleration $a = a_0 (1 - t/T)$ where a_0 and T are constants. The particle at t = 0 has zero velocity. Calculate the average velocity between t = 0 and the instant when a = 0.

Q15. Two trains A and B 100 m and 60 m long are moving in opposite directions on the parallel tracks. The speed of shorter train is 3 times that of the longer one. If the train take 4 seconds to cross each other then find the velocities of the trains.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	C	A	B	A	A	A	D	B	A
11	12	13	14	15	16	17	18	19	20
C	D	C	C	A	C	D	A	B	C
21	22	23	24	25	26	27	28	29	30
B	C	B	C	C	C	C	D	A	B
31	32	33	34	35	36	37	38		
B	C	A	D	C	B	D	A		

Ex - 1A

1. B, D **2. A, C** **3. A, B** **4. A, C** **5. A, B, C**

6. A, B, C **7. A, B, D** **8. A, D** **9. A, C, D** **10. A, B, C, D**

11. A, B

12. A→q; B→p,q; C→p,r; D→r,s

13. A→p; B→v; C→s; D→q; E→u; F→t; (G)– r

14. A→r; B→q; C→s; D→p; **15. D** **16. D** **17. A**

18. 18.5 m **19. 89 m** **20. 9**

21. (i) 20 m, (ii) 5 s

Ex-2										
1	2	3	4	5	6	7	8	9	10	11
D	C	A	C	A	A	A	D	B	A	C
12	13	14	15	16	17	18	19	20	21	
A	A	A	D	B	D	B	C	D	C	

Ex - 2A

1. C, D **2.** B,C,D **3.** A, C **4.** A,C,D **5.** C, D

6. D **7.** B **8.** A **9.** B **10.** B

11. A→r; B→p; C→s; D→q **12.** 12 km

13. (a) $\sqrt{60}$ m/s, (b) $\sqrt{\frac{3}{5}}$ sec, (c) $\frac{3}{\sqrt{60}}$ sec, (d) $\frac{9}{4}$ m

14. $\frac{a_0 T}{3}$

15. $V_A = 10$ m/s , $V_B = 30$ m/s